STATEMENT OF WORK

1.0 Introduction/Background: The NASA Langley Research Center Hypersonic Materials Environmental Test System (HYMETS) is a 400 kW arcjet facility that is looking to purchase new fused silica optical viewports to replace existing fused silica optical viewports present in the facility. While the current viewports are still adequate and serve their purpose, replacement viewports are necessary as the optical properties of these viewports degrade over time with use, and the current windows are approximately 10 years old. These viewports have significant lead times, so starting the process now when funding is available ensures these viewports can be utilized sooner in the facility to meet customer requirements. These optical viewports are the primary means of performing optical diagnostics within the HYMETS facility, and are critical to customer test entries and data.

2.0 Scope of Work: The scope of this statement of work covers the objectives, work description, and deliverables necessary for the purchase of new fused silica optical viewports for the HYMETS facility.

3.0 Objectives: The objective of this statement of work is to obtain new fused silica optical viewports for the HYMETS facility to replace the current viewports which are old and degrading.

4.0 Description of the Work/Contractor Tasks: The work described in this statement of work is as follows per the provided quote from the vendor. The vendor:

1) Shall provide a quantity of 3 Low OH Infrared 2.88-in. fused silica optical viewports with a Single Layer Anti-Reflection (SLAR) coating from 700nm to 2200nm in their standard weldable configuration. (Line Item 0)

2) Shall then weld a threaded adapter each to the items specified in Item 1 above, per the vendor provided drawing. (Line Item 1)

3) Shall provide a quantity of 4 DUV200 2.88-in. fused silica optical viewports, no anti-reflection coating, in their standard weldable configuration. (Line Item 2)

4) Shall then weld a threaded adapter each to the items specified in Item 3 above, per the vendor provided drawing. (Line Item 3)

5) Shall provide a quantity of 5 EUV185 Zerol 2.50-in. fused silica optical viewports, no anti-reflection coating, in their standard 4.50-in. CF configuration. (Line Item 4)

6) Shall provide a quantity of 3 UV GRD Zerol 2.50-in. fused silica optical viewports, no anti-reflection coating, in their standard 4.50-in. CF configuration. (Line Item 5)

5.0 Deliverables: The vendor shall deliver the following items and quantities per the schedule provided in their quote:

1) Qty = 3, Low OH Infrared 2.88-in. fused silica optical viewport with SLAR coating from 700nm to 2200nm in a threaded weldable configuration per the vendor provided drawing. (16 weeks after final print approval)
2) Qty = 4, DUV200 2.88-in. fused silica optical viewport, no anti-reflection coating, in a threaded weldable configuration per the vendor provided drawing. (12 weeks after final print approval)
3) Qty = 5, EUV185 Zerol 2.50-in. fused silica optical viewport, no anti-reflection coating, in their standard 4.50-in. CF configuration. (12 weeks ARO)
4) Qty = 3, UV GRD Zerol 2.50-in. fused silica optical viewport, no anti-reflection coating, in their standard 4.50-in. CF configuration. (12 weeks ARO)

Other Considerations:

Applicable Documents/Background: Insulator Seal, now MDC Precision, LLC, and a contractor, PEC Professionals, created threaded weldable fused silica optical viewports for the HYMETS facility. The viewports were a combination of Insulator Seal's COTS weldable fused silica optical viewports and a threaded adapter provided by PEC Professionals that interfaced with the threaded ports of the HYMETS facility. Insulator Seal created a unique drawing specific to these parts, Drawing Number 1201601. PR NNL15AD23P was created and used to purchase these viewports back in 2015. After a decade of significant use, these viewports are now due for replacement. Unfortunately PEC Professionals is no longer in business. MDC Precision was contacted and asked if they could still manufacture these viewports per their original drawing. MDC Precision replied yes, and provided a quote. Please note that only Items 0 & 1 and Items 2 & 3 in the quote are for the custom threaded weldable fused silica optical viewports. Items 4 & 5 in the quote are unmodified MDC Precision COTS viewports that could be purchased from other vendors, but are bundled in this quote to save time and effort by not duplicating this process since a PR would be required to purchase them as well and those viewports also have their own manufacturing lead times. All items are needed and will be used on the HYMETS facility. To the best of my understanding, another vendor cannot be used for these parts since they were created based on a proprietary drawing owned by MDC Precision and use MDC precision COTS parts. However, another custom optical viewport vendor, Rayotek Scientific, LLC was contacted and asked if they could duplicate the threaded weldable fused silica optical viewports, and their response was "…cannot economically copy what was done before" and "…have to no-bid this. Its no in our wheelhouse [sic]".

Milestones: Purchase of these fused silica optical viewports uses Reimbursable Space Act Agreement funds (SAA1-41034) which were provided as payment for services rendered the week of February 3rd, 2025 in the HYMETS facility. These funds are due to expire (be returned to the customer) on 03/28/2025. Funds NEED to be committed before they expire.